

GOVERNANCE_AUDIT_2026-08-07

Governance & Constitution Compliance Audit — 2026-08-07

Revision: 1 **Last modified:** 2026-08-07T19:10:54Z **Status:** active **Scope:** Full-repository audit against the inherited Helix Constitution (constitution/CLAUDE.md, constitution/Constitution.md) + verification of docs/REMAINING_WORK_PLAN.md (2026-06-14) closure claims + live-toolchain health. Assembled from 6 parallel read-only subagent audits, 2026-08-07. No file was modified by the audit itself; every finding below cites file:line, commit, or command output as evidence (§11.4.6 — no guessing).

Baseline: HEAD 0d05ec1 on main. docs/CONTINUATION.md was last updated at commit 646b295 (2026-06-16, “Session 14”) — 21 commits / ~52 days behind HEAD at audit time.

0. How to execute this plan

Same discipline as REMAINING_WORK_PLAN.md: each GA-NN item is TDD-first (RED before fix, §11.4.43/§11.4.115), gets a regression guard where it closes a defect (§11.4.135), and closes only with pasted live command output (no self-certification words without evidence). Items marked **OPERATOR-DECISION** are surfaced, not auto-executed. Non-contending items run in parallel subagent streams (§11.4.70/§11.4.103).

Phase 0 — Toolchain unblock (P0, prerequisite for verifying everything else)

GA-01 — Python test toolchain is ABI-broken; no project .venv

- **Evidence:** live run of the documented command
`python3 -m pytest tests/unit/ -q --import-mode=importlib`
crashes during plugin auto-load: `ModuleNotFoundError: No module named 'rpds.rpds' / No module named 'pydantic_core._pydantic_core'`. `python3` resolves to 3.14.6; every native extension in `~/.local/lib/python3/site-packages` is compiled for `cpython-313`. No `.venv/` exists in the repo. **Zero tests can currently run** — `CONTINUATION.md`'s "4277 passed" claim is unreproducible today.
- **Blast radius:** blocks the entire unit suite, RW-21 verification, and 3 Challenge scripts (`download_proxy_deep_challenge.sh`, `private_tracker_html_challenge.sh`, `search_deep_coverage_challenge.sh`) that need `.venv`.
- **Fix:** create a project-pinned `.venv` (Python 3.12, matching `pyproject.toml`'s `target-version/mypy python_version`), `pip install -r tests/requirements.txt` (+ any download-proxy requirements), verify `pytest tests/unit/ -q --import-mode=importlib` actually collects and runs, paste the real N-passed/N-failed count. Document the exact bootstrap in `docs/TESTING.md` and `CONTINUATION.md`'s Quick-Start (which currently omits `venv` activation entirely).
- **Acceptance:** real `pytest` run completes (not crashes) with a captured summary line; count compared against the 4277 baseline and any delta explained.
- **Priority:** P0.

GA-02 — Frontend Vitest toolchain broken (stale node_modules)

- **Evidence:** `cd frontend && npx vitest run` fails before any test executes: `Error: Cannot find module '@analogjs/vite-plugin-angular' while loading vitest.config.ts`. `frontend/node_modules` exists (332M) but is incomplete/stale.
- **Fix:** `cd frontend && npm ci` (or `npm install` if no lockfile match), re-run `npx vitest run`, capture the real pass count.
- **Priority:** P0.

GA-03 — Extension Vitest toolchain never bootstrapped

- **Evidence:** extension/node_modules absent entirely. All 8 challenges/extension/*.sh scripts fail cleanly and honestly (FAIL: vitest not installed ... run 'cd extension && npm install' first) rather than false-passing — the challenges themselves are not buggy, the environment is just never set up. Blocks verification of the “379 passed / 33 spec files” claim in CONTINUATION.md Session 10.
 - **Fix:** cd extension && npm install, re-run npx vitest run + all 8 extension challenges, capture real counts.
 - **Priority: P1.**
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Phase 1 — Governance/tracking seam reconnection

Root cause: the mandated tracking chain (CONTINUATION.md → Issues.md/Fixed.md/workable_items.db → README entry-point) stopped being touched on 2026-06-16 while real engineering work (19 substantive commits) kept landing cleanly and correctly on main. The code is fine; the seam that is supposed to keep governance docs synced with code was silently disconnected — precisely the “prose exists, nobody wired the seam” failure mode the constitution’s own §11.4.205/ §11.4.226/ §11.4.227 anchors describe.

GA-04 — CONTINUATION.md 52-day / 21-commit staleness gap

- **Evidence:** git log --oneline 646b295..HEAD = 21 commits (19 substantive), spanning 2026-06-16T23:30 → 2026-08-07T21:52, none reflected in the file. Themed: Cyrillic/UTF-8 encoding-fix wave (6 commits, 029c045..cdec407), tracker/plugin fixes (7), egress/proxy/Jackett feature wave (4: 2f160cd,4e62eca,b44a2e4,be5062d), doc-sync (1), host-mirror sync (2, no-op for CONTINUATION purposes).
- **Violates:** §12.10 / §11.4.131 (standing session-resumption file, always current).
- **Fix:** author a new “Session 15” entry summarizing all 19 substantive commits by theme, update **Last commit:** **Last modified:** **Revision:**, reconcile the “In flight — NEXT ITEM” section (the P0 Cyrillic item it names as in-flight is now DONE per commit 4aee3c4/8918d4b — verify and close it in the doc).
- **Priority: P0.**

GA-05 — Lava P1–P4 porting work never landed as tracked workable items

- **Evidence:** docs/PORTING-FROM-LAVA.md (created be5062d, 2026-07-01) maps 4 items to Lava-audit finding numbers: P1 §5 durable remote-exec, P2 §0/§4 egress/VPN routing, P3 §3 configurable outbound proxy, P4 §8 Jackett cookie-session login. All four are implemented in code (scripts/deploy-remote.sh, containers/pkg/remote-executor.go, scripts/egress-via-vpn.sh, download-proxy/src/config/proxy.py, qBitTorrent-go/internal/httpx/proxy.go, commit 2f160cd explicitly “port Lava §8 finding”) — but `grep -in "lava|egress|BOBA_UPSTREAM_PROXY|durable-run|remote-exec" docs/Issues.md docs/Fixed.md` returns zero hits. A §6/§9 item (“anti-bluff QA verdict/diagnostics strengthening”) is listed applicable in the porting doc’s own triage table but no commit closes it — status unknown.
- **Violates:** §11.4.148 / §11.4.202 / §11.4.208 (every actionable finding must land as a tracked item, not merely a commit message).
- **Fix:** create BOB-064..067 via the workable-items tool (not raw MD edits, to keep DB↔MD custody intact — `workable-items validate/diff` already confirmed the DB↔MD pair is internally in sync, just frozen at BOB-063), cite the implementing commit + file as evidence, close as Implemented/Fixed. Determine the true status of the §6/§9 item and either close it with evidence or track it honestly as open.
- **Priority:** P0.

GA-06 — docs/PORTING-FROM-LAVA.md missing §11.4.44 revision header

- **Evidence:** `grep -n Revision docs/PORTING-FROM-LAVA.md` → empty.
- **Fix:** add `**Revision:** N / **Last modified:**` header block.
- **Priority:** P2.

GA-07 — README.md has no §11.4.57/§11.4.212 doc-link entry-point section

- **Evidence:** README.md’s ## Documentation section (lines 108–162) links 20+ docs but contains **zero** references to docs/Issues.md, docs/Fixed.md, docs/CONTINUATION.md, docs/PORTING-FROM-LAVA.md, docs/browser_extension/Status.md, docs/browser_extension/RELEASE_READINESS.md, docs/COMPLETION_STATUS.md, or docs/

RELEASE_READINESS_20260616.md. Every tracker/status doc audited here is an orphan relative to README.

- **Violates:** §11.4.57 (Tracked-Items + Status Documents mandated section) / §11.4.212 (README as canonical entry point for ALL project documentation).
- **Fix:** add the mandated table (Document | Last modified | Revision | Markdown | HTML | PDF) covering every tracker/status doc in the repo, including this audit doc and REMAINING_WORK_PLAN.md.
- **Priority: P0** (structural, repo-wide).

GA-08 — docs/browser_extension/Status.md stale by 4 commits

- **Evidence:** Rev 15, 2026-06-13T13:10:00Z. Commits since, untouched in the doc: a410b91 (coverage), 0897248 (video-confirm), ee4a01b (crash-hardening), 0dc247b (qBittorrent 5.x compat).
- **Fix:** update Status.md + regenerate Status_Summary.md + HTML/PDF siblings (§11.4.45/§11.4.56).
- **Priority: P2.**

GA-09 — workable_items.db: BOB-008 missing operator_block_details row

- **Evidence:** `workable-items validate --db docs/workable_items.db` → 1 violation: “BOB-008: Operator-blocked with no operator_block_details row (§11.4.148 D3)”. The prose in Issues.md DOES contain the `**Operator-Block-Details:**` line — this is a DB-column population gap, not a doc gap.
- **Fix:** populate via the `workable-items` tool’s proper subcommand (not raw SQL).
- **Priority: P3.**

GA-10 — No successor to docs/RELEASE_READINESS_20260616.md; no top-level v1.0.0 ledger

- **Evidence:** the doc is a self-labeled point-in-time snapshot (HEAD 8b9ef90); 20 commits since have touched the audited surfaces. Its own text names “no top-level proxy/merge-service v1.0.0 readiness ledger” as an open blocker — this is RW-20 in the existing plan, still NOT-DONE (confirmed by the RW-plan verification agent).
- **Fix:** create the ledger doc (dedupes with RW-20 — track under one ticket, not two).

- **Priority: P2.**
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Phase 2 — Security correctness gaps (corrections to RW-plan closure claims)

GA-11 — RW-02 incomplete: 2 mutating routes still unauthenticated

- **Evidence:** `routes.py/scheduler.py`'s auth rollout covers download/upload/file/magnet (`routes.py`) and POST/DELETE schedules — but PATCH `/api/v1/schedules/{id}` (`scheduler.py:108`) and PUT `/api/v1/theme` (`routes.py:82`) have **no** `Depends(require_api_token)`. Confirmed by omission: `tests/security/test_hooks_schedules_auth.py:194-198`'s own `_MUTATING` enumeration never included them — the original fix and its test were scoped incompletely together.
- **Fix (TDD):** RED — extend `_MUTATING` to include PATCH schedule + PUT theme, confirm the new assertions fail (401 expected, currently unauthenticated) → add `Depends(require_api_token)` to both routes → GREEN → live curl-verify 401-unauth / 200-with-token on a running container.
- **Priority: P0** (security).

GA-12 — RW-06 unverifiable: rutracker ReDoS fix not confirmed deployed

- **Evidence:** source fix present (`plugins/rutracker.py:140, {0,512}` bound); no live container running on the audit host to check `config/qBittorrent/nova3/engines/rutracker.py` (the installed copy) or measure live parse timing — a \$11.4.108 SOURCE=ARTIFACT/RUNTIME gap that is unverified, not necessarily unfixed.
- **Fix:** bring up the compose stack (`./start.sh`), run `./install-plugin.sh`, verify `grep '{0,512}' config/qBittorrent/nova3/engines/rutracker.py` on the container, capture timing of a large rutracker result page.
- **Priority: P1** (needs live stack).

GA-13 — RW-07 partial: nnmclub e2e SKIP-on-404 fallback never removed

- **Evidence:** tests/e2e/test_live_stack_evidence.py:265 still contains the skip fallback; the acceptance criterion (“un-skip”) is unmet regardless of live container state — the route existing in source (download-proxy/src/api/auth.py:350) isn’t what was asked for.
 - **Fix:** with the live stack up, confirm `curl :7187/api/v1/auth/nnmclub/status → 200`, remove the skip fallback, let the assertion run for real, capture the pass.
 - **Priority: P1** (needs live stack).
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Phase 3 — Test-type integrity (constitution’s “non-unit tests hit real services” rule)

GA-14, GA-15, GA-16 — Three test files mislabeled by directory mock the exact system under test

- **Evidence:**
 - tests/integration/test_merge_api.py (577 lines) — own docstring: “Uses FastAPI TestClient with mocked SearchOrchestrator.” Every route test `@patch("api.routes._get_orchestrator")`.
 - tests/e2e/test_full_pipeline.py — docstring claims “End-to-end”, but patches `SearchOrchestrator._get_enabled_trackers/_search_tracker` with a synthetic fake — no real tracker HTTP call ever happens.
 - tests/contract/test_tracker_stats_contract.py:80-81 — same `monkeypatch.setattr` pattern on the same two methods.
- **Violates:** CLAUDE.md’s inherited rule: “Mocks, stubs... are permitted ONLY in unit tests ... All other test types MUST interact with real fully implemented System” (§11.4.27).
- **Fix:** for each file — (a) relocate the mocked version into tests/unit/ under an honest name (the coverage IS valuable as unit-level route-contract testing), AND (b) author a new real-service replacement in the original directory using the project’s EXISTING live-stack fixtures (tests/fixtures/compose.py, tests/fixtures/services.py, tests/integration/test_fixtures_bring_up_services.py already demonstrate this

pattern is viable in this repo) so the directory's contract (integration/e2e/contract = real services) is actually met.

- **Priority: P1** (each file independent, parallelizable).

GA-17 — #legacy-untriaged skip-tag family never actually triaged

- **Evidence:** ~10 sites (tests/e2e/test_public_trackers_return_results.py:128, tests/unit/test_openapi_frozen.py:15,23, tests/unit/test_plugin_smoke.py:45, tests/docs/test_no_broken_links.py:51, tests/fixtures/services.py:9, others) carry SKIP-OK: #legacy-untriaged — syntactically satisfies the format, but the “ticket” is a literal placeholder admitting it was never triaged.
- **Fix:** triage each site individually — determine the real reason, either fix/un-skip or replace the placeholder with a real ticket id + reason.
- **Priority: P3.**

GA-18 — 7 permanently-skipped dead Jaxtt-autoconfig Python tests

- **Evidence:** tests/contract/test_jaxtt_autoconfig_contract.py:18,24, tests/integration/test_jaxtt_autoconfig_real.py:71,77,83, tests/e2e/test_jaxtt_autoconfig_e2e.py:32,38, tests/security/test_jaxtt_autoconfig_secrets.py:41 — all skip with “endpoint moved to boba-jaxtt:7189”. Migration confirmed deliberate: download-proxy/src/api/__init__.py:301-306 explicitly removes the Python endpoint in favour of the Go service (RW-13 evidence).
 - **Fix (§11.4.124 investigate-before-remove — investigation above already satisfies the proof requirement):** delete these dead test bodies in one dedicated commit citing the git-history/migration evidence; the replacement coverage lives in qBitTorrent-go/tests/integration/jaxtt_db_test.go and the Go autoconfig package.
 - **Priority: P3.**
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Phase 4 — Go backend parity (blocked on operator decision)

GA-19 — OPERATOR-DECISION: is --profile go parity still a release goal?

- **Evidence:** this is RW-09, unresolved since 2026-06-14. docs/migration/PARITY_GAPS.md and docs/features/Status.md:52,194 are unrevised (“Last audited 2026-04-27”, 6 ported / 4 partial / 8 missing). Confirmed-missing since the plan was written: scheduler driver loop (qBitTorrent-go scheduler files have no time.Ticker), metadata enricher (no equivalent package), public-tracker plugin fan-out (no exec.Command fan-out in internal/service). Jackett autoconfig (RW-13) is now PARTIAL — implemented as its own Go service, canonical per download-proxy/src/api/__init__.py:301-306, but BEP48/scrape validation and a CAPTCHA REST endpoint are still absent from the Go side.
- **This gates:** RW-10 (scheduler driver — “highest-impact silent functional hole on the Go path” per the RW-plan’s own words), RW-11 (enricher), RW-12 (plugin fan-out), the remainder of RW-13, and GA-20’s stress/chaos items RW-14/RW-15 to the extent they’re Go-path-specific.
- **Decision needed:** commit real scope to close the 3 missing Go subsystems (multi-day effort), or mark the Go backend a documented future blueprint and stop implying it’s a near-term parity target in Status.md.
- **Priority: P1, surfaced via §11.4.66 — not auto-executed.**

GA-20 — RW-14/RW-15 stress/chaos coverage claims were inaccurate; still NOT-DONE

- **Evidence:** commit a410b91 (“close Go + extension coverage gaps RW-14/RW-16”) actually added Go internal/jackett/internal/logging unit tests and an extension parser test — NOT §11.4.85-conformant stress/chaos coverage for the tracker-fetch/cookie-auth path (RW-14) or the scheduler+hooks+SSE-broker surface (RW-15). No matching test files exist. RW-16’s Go-DB concurrent-write test (jackett_db_test.go:440-500) predates the plan (commit 8936545, 2026-04-27) and has no fault-injection — genuinely PARTIAL, not the chaos coverage the plan calls for.
- **Fix:** author real tests/stress/test_tracker_fetch_stress_chaos.py (download-proxy cookie-auth path, overlapping the RW-03 SSRF

surface) and tests/stress/
test_scheduler_hooks_sse_stress_chaos.py, plus a chaos-fault-
injection extension to the existing Go DB concurrency test (process-
kill / resource-exhaustion, not just concurrent writes). Correct the
historical claim in a follow-up doc note (§11.4.6 — don't leave the
false “closed” claim standing).

- **Priority: P2.**
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Phase 5 — Challenges / HelixQA infrastructure bugs

GA-21 — jackett_autoconfig_clean_slate.sh polls a retired endpoint

- **Evidence:** 3 references to /api/v1/jackett/autoconfig/last (lines 8, 58, 62) — this endpoint was deliberately removed per CLAUDE.md (“The Python /api/v1/jackett/autoconfig/last endpoint was removed”). The challenge would always fail step 5 against the current architecture.
- **Fix:** update to poll the Go :7189 autoconfig-runs API (GET /api/v1/jackett/autoconfig/runs OR /runs/{id} per qBitTorrent-go/internal/jackettapi/router.go).
- **Priority: P1.**

GA-22 — All 6 HelixQA bank symlinks are dangling absolute macOS paths

- **Evidence:** challenges/helixqa-banks/*.yaml → symlinks to /Volumes/T7/Projects/Boba/submodules/helixqa/banks/*.yaml, authored on a different dev machine. Broken on this (and any non-that-exact-Mac) host. The real files exist and are correct at submodules/helixqa/banks/*.yaml (37-entry populated submodule, confirmed not empty).
- **Fix:** replace with relative symlinks (ln -sf ../../submodules/helixqa/banks/<name>.yaml).
- **Priority: P1.**

GA-23 — HelixQA bank action steps hardcode /Volumes/T7/Projects/Boba/...

- **Evidence:** e.g. boba-boba-ctl.yaml: `cd /Volumes/T7/Projects/Boba/cmd/boba-ctl && go build ...`; boba-docs-chain.yaml: `bash /Volumes/T7/Projects/Boba/scripts/docs_chain.sh`. Non-portable to this or any other checkout.
- **Fix:** parametrize with a `$PROJECT_ROOT/$REPO_ROOT` variable the HelixQA runner resolves at invocation time, or make paths script-relative.
- **Priority:** P2.

GA-24 — no_suspend_calls_challenge.sh false-positives on a governance doc

- **Evidence:** flags `constitution/docs/scripts/guard-forbidden-commands.md` because that doc *quotes* the forbidden `systemctl/loginctl` verbs it documents/enforces. Its `EXCLUDE_PATHS` allowlist covers `CONSTITUTION.md/AGENTS.md/CLAUDE.md` but not this file — a carrier false-positive, the exact §11.4.201 pattern the constitution itself warns about.
 - **Fix:** add the file to `EXCLUDE_PATHS` in `scripts/host-power-management/check-no-suspend-calls.sh`.
 - **Priority:** P3.
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Phase 6 — CONST-033 real signal (HIGH — project's own hard-ban class)

GA-25 — host_no_auto_poweroff_challenge.sh found 1 genuine post-fix poweroff broadcast

- **Evidence:** 4/5 checks pass; 1 fails — a “will power off” journal broadcast recorded AFTER the documented fix marker (2026-04-28T20:54:31+03:00).
- **This is the exact incident class CLAUDE.md's own CONST-033 Operational Note exists for.** Per that note, the correct response is triage BEFORE assuming host-code causation and BEFORE dismissing it: (1) uptime cross-check against any alleged downtime window, (2) `journalctl -k --since "24 hours ago" | grep -iE "will suspend|systemd-suspend"`, (3) `journalctl -k --since "24 hours ago" | grep -iE "oom-kill|killed process"` + decode the

oom_memcg cgroup path if present, (4) re-run both CONST-033 challenges, (5) document findings in docs/incidents/<date>-* .md regardless of outcome. **Only read-only diagnostic commands** — nothing that triggers a power-state transition, per the absolute Hard Ban.

- **Priority: P0** — triage this first, independent of everything else in this plan, given the project's own stated severity for this incident class (prior data-loss history cited in CLAUDE.md).
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Phase 7 — Gate-scoping bug

GA-26 — `ruff check . sweeps submodules/containers/` (a separately-governed submodule)

- **Evidence:** `[tool.ruff]` in `pyproject.toml` has no `exclude/extend-exclude` at all. 71 `ruff` violations found; sampled violations are exclusively in `submodules/containers/scripts/resource-policy/*.py` — a submodule that owns its own engineering standards per §11.4.28 decoupling. This inflates boba's own lint-gate count with debt that isn't boba's to fix.
 - **Fix:** add `extend-exclude = ["submodules/", "constitution/", ".worktrees/"]` to `[tool.ruff]`, re-run `ruff check .`, report the TRUE boba-only violation count (may be 0, may be nonzero — verify, don't assume).
 - **Priority: P2.**
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Phase 8 — Governance-layer ambiguity (needs operator clarification)

GA-27 — OPERATOR-DECISION: Hard Stop #3 vs. this project's documented maintenance workflow

- **Evidence:** the inherited constitution's Hard Stop #3 states container orchestration must be owned exclusively by the project's own binary/orchestrator, with direct `docker/podman` commands "prohibited as workflows." This project's own CLAUDE.md "Critical Constraints" section explicitly prescribes `direct podman exec qbittorrent-proxy ...`, `podman restart qbittorrent-proxy`, and `podman compose down && podman compose up -d` as the mandated

procedure for Python-source reload, plugin reload, and compose/env changes.

- **Decision needed:** either (a) formally classify this as an intentional, documented exception per §11.4.17 (with a one-line rationale — e.g. “restart-level granularity below full stack lifecycle is exempt because start.sh/stop.sh don’t yet expose it”), or (b) extend start.sh/stop.sh with subcommands that wrap these exact operations (e.g. ./start.sh --restart-proxy, ./start.sh --reload-plugins) so operators never type raw podman commands, closing the gap for real.
- **Priority: P1, surfaced via §11.4.66 — not auto-executed.**

Summary table

Phase	Items	Blocking on live infra?	Blocking on operator decision?
0 — Toolchain unblock	GA-01..03	No	No
1 — Governance/tracking	GA-04..10	No	No
2 — Security corrections	GA-11..13	GA-12/13 yes	No
3 — Test-type integrity	GA-14..18	GA-14..16 partial (real-service replacements)	No
4 — Go parity	GA-19..20	No	GA-19 yes
5 — Challenge/HelixQA bugs	GA-21..24	No	No
6 — CONST-033 signal	GA-25	No (read-only triage)	No
7 — Gate scoping	GA-26	No	No
8 — Governance ambiguity	GA-27	No	GA-27 yes

Cross-reference: GA-11..13, GA-19..20 correct/extend docs/
REMAINING_WORK_PLAN.md's RW-02/06/07/09/14/15 entries — that
document's status table should be updated in the same pass to avoid
two divergent trackers describing the same defects (§11.4.186 anti-
divergence).

Definition of done for this audit: every GA item above reaches DONE
(with live-captured evidence), OPERATOR-DECISION items (GA-19,
GA-27) are resolved, REMAINING_WORK_PLAN.md's corrected items are
reconciled, README carries the doc-link section, CONTINUATION.md is
current, and every fix ships with a regression guard + (where the
surface is user-facing) a HelixQA Challenge entry.